

9 I is inversely proportional to the square of r .

(a) When $r = \frac{1}{5}$, $I = 500$. Find I when $r = \frac{1}{10}$.

Answer _____ [2]

(b) When the value of r is tripled, find the percentage decrease in the value of I .

Answer _____ % [2]

- 10 Find **two pairs of numbers**, all numbers being smaller than 1400, that have a lowest common multiple of 5400 and a highest common factor of 90.

Answer _____ and _____ [3]

- 11 (a) Factorise $4a^2 + 12ab + 9b^2$

Answer _____ [1]

- (b) Hence, completely factorise $4a^2 + 12ab + 9b^2 - 1$

Answer _____ [2]

(c) n is a positive integer.

Show that, for all n , $(3n+1)^2 - (3n-1)^2$ is a multiple of 4.

[2]

12 Sketch the graph of $y = -(x+3)^2 + 5$ on the axes below.

Indicate clearly the coordinates of the points where the graph crosses the axes and the turning point on the curve. [3]

